Kruskal-EDS: Edge Dynamic Stratification

A Distribution-Adaptive Minimum Spanning Tree Algorithm Inspired by Ergodic Theory

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Abstract

We introduce **Kruskal-EDS** (*Edge Dynamic Stratification*), a distribution-adaptive variant of Kruskal's minimum spanning tree (MST) algorithm that replaces the mandatory $\Theta(m \log m)$ global sort with a three-phase procedure inspired by Birkhoff's ergodic theorem.

In Phase 1, a sample of $\sqrt{m}$ edges estimates the weight distribution in $O(\sqrt{m} \log m)$ time. In Phase 2, all m edges are assigned to k strata in $O(m \log k)$ time via binary search on quantile boundaries — no global sort. In Phase 3, strata are sorted and processed in order; execution halts as soon as $n-1$ MST edges are accepted.

We prove an effective complexity of $O(m + p \cdot (m/k) \log(m/k))$, where $p \leq k$ is the number of strata actually processed. On sparse graphs or heavy-tailed weight distributions, $p \ll k$ and the algorithm achieves near-$O(m)$ behaviour. We further derive the optimal strata count $k^* = \lceil \sqrt{m/\ln(m+1)} \rceil$, balancing partition overhead against intra-stratum sort cost.

An extensive benchmark on 14 graph families demonstrates correctness on 12 test cases and practical speedups reaching **10**$\times$ in wall-clock time and **33**$\times$ in sort operations over standard Kruskal. A 3-dimensional TikZ visualisation of the complexity landscape illustrates the algorithm's adaptive behaviour as a function of graph density and weight distribution skewness.

Keywords: minimum spanning tree, Kruskal's algorithm, distribution-adaptive sorting, stratified sampling, Birkhoff ergodic theorem, edge partitioning, sparse graphs.

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1 Introduction

The *Minimum Spanning Tree* (MST) problem is one of the most fundamental in combinatorial optimisation. Given a connected weighted graph $G = (V, E, w)$ with $|V| = n$ vertices and $|E| = m$ edges, an MST is a spanning tree of minimum total weight. MST algorithms underlie applications in network design, image segmentation, clustering, VLSI layout, and approximation algorithms for NP-hard problems such as the Travelling Salesman Problem [Cormen et al., 2022].

The sorting bottleneck. Kruskal's algorithm [Kruskal, 1956] remains one of the simplest and most widely deployed MST algorithms: *sort all edges by weight, then greedily add the lightest edge that does not form a cycle.* Its complexity is dominated by the global sort: $\Theta(m \log m)$. This is optimal among comparison-based sorts on unstructured inputs. However, *the algorithm uses only $n-1$ edges out of m*, and on sparse graphs or graphs with favourable weight distributions, sorting all m edges is wasteful.

The ergodic insight. In dynamical systems, Birkhoff's ergodic theorem [Birkhoff, 1931] guarantees that time averages along almost every orbit converge to the space average — a small sample suffices to characterise the global measure. Artur Ávila's Fields Medal work on quasi-periodic cocycles [Ávila, 2015] pushes this further: the Lyapunov spectrum of a linear cocycle is determined by its behaviour on an exponentially small set of orbits.

We draw an analogy: the weight distribution of E is the *invariant measure* of the MST problem. A sample of $\sqrt{m}$ edges estimates this measure with $O(1/\sqrt{m})$ quantile error (by the Dvoretzky-Kiefer-Wolfowitz inequality [Dvoretzky, Kiefer, and Wolfowitz, 1956]). From this estimate, we construct k *strata* (weight intervals) that partition E in $O(m \log k)$ time without any global sort, then process strata in order until the MST is complete.

Contributions.

C1. Algorithm: Kruskal-EDS, a three-phase MST algorithm with distribution-adaptive complexity (§4).

C2. Theory: proof of correctness, complexity analysis with optimal strata count k^*, and analysis on specific distributions (§5).

C3. 3D Complexity Landscape: a novel TikZ visualisation of the algorithm's complexity as a function of density and distribution skewness (§6).

C4. Empirical evaluation: benchmark on 14 graph families across four weight distributions (§6).

2 Background and Related Work

2.1 Classical MST Algorithms

Kruskal (1956). Sort edges by weight, then apply a Union-Find structure to test and merge components [Kruskal, 1956]. Total complexity: $\Theta(m \log m)$ (sort-dominated).

Prim (1957). Grow the MST from a seed vertex using a priority queue. With a Fibonacci heap: $O(m + n \log n)$ [Prim, 1957]. Superior when $m = \omega(n \log n)$, i.e. on dense graphs.

Borůvka (1926). In each phase, each component selects its lightest outgoing edge. After $\log n$ phases: $O(m \log n)$ [Borůvka, 1926]. Naturally parallelisable.

Fredman-Tarjan (1987). Combines Borůvka phases with Fibonacci-heap Prim: $O(m\,\beta(m,n))$ where $\beta(m,n) = \min\{i : \log^{(i)} n \leq m/n\}$ [Fredman and Tarjan, 1987].

Chazelle (2000). The first near-linear MST algorithm: $O(m\,\alpha(m,n))$ using soft heaps [Chazelle, 2000], where α is the inverse Ackermann function.

Linear MST (randomised). Karger, Klein, and Tarjan (1995) achieved expected $O(m)$ via random sampling and a recursive verification procedure [Karger, Klein, and Tarjan, 1995]. This is asymptotically optimal but complex to implement.

Optimal deterministic MST. The decision-tree complexity of MST is $\Theta(m)$ [Pettie and Ramachandran, 2002], but no explicit $O(m)$ deterministic algorithm is known.

2.2 Distribution-Aware Sorting and Partitioning

Integer sorting (radix sort, counting sort) achieves $O(m)$ for bounded integer keys [Knuth, 1998]. Sample sort [Frazer and McKellar, 1970] partitions m elements using $\sqrt{m}$ splitters, sorting each bucket independently. Our stratification generalises sample sort to the MST context with *early termination*: strata beyond those needed for the MST are never sorted.

Thorssen and Kaplan [2002] study MST algorithms exploiting integer weight distributions; our approach is complementary, targeting real-valued weights with arbitrary distributions.

2.3 From Ergodic Sampling to Edge Stratification

The efficiency of Kruskal-EDS relies on the assumption that a sub-sampling of size $s \ll m$ captures the essential geometry of the weight distribution. Formally, we view the set of edges E as a discrete probability space $(E, \mathcal{F}, \mu)$, where μ represents the empirical distribution of weights $w : E \to \mathbb{R}^+$.

In the spirit of *Birkhoff's Ergodic Theorem*, we treat the random sampling of s edges as an observation of the system's "invariant measure". If we consider the graph construction as a process where weights are drawn from a continuous density $f(x)$, the sampling phase aims to reconstruct the Lyapunov-like growth of the MST.

Following the works of Avila on the quantization of measures, our stratification can be seen as an optimal partition of the weight space. The DKW inequality (Equation **??**) then provides the speed of convergence: it ensures that the empirical measure $\hat{\mu}_s$ obtained by our sample deviates from the true measure μ by no more than ε with high probability. This bridge justifies why a "local" view (the sample) is sufficient to dictate a "global" strategy (the partitioning of m edges).

2.4 Ergodic Theory and Algorithms

The connection between ergodic theory and algorithms is well-established in mixing time analysis of Markov chains [Levin and Peres, 2017] and pseudo-random generator construction. Ávila's work on quasi-periodic cocycles [Ávila, 2015] establishes that a small orbit determines

the global Lyapunov spectrum — a quantitative instance of the ergodic principle. To our knowledge, the application of Birkhoff sampling to adaptive edge stratification for MST has not been previously studied.

3 Problem Statement and Notation

Throughout, $G = (V, E, w)$ denotes a connected undirected weighted graph with $n = |V|$ vertices and $m = |E|$ edges, $w : E \to \mathbb{R}$. We write w_e for the weight of edge e.

Definition 3.1 (Minimum Spanning Tree)**.** A spanning tree $T \subseteq E$ with $|T| = n-1$ is a *minimum spanning tree* if $w(T) := \sum_{e \in T} w_e$ is minimised over all spanning trees of G.

Definition 3.2 (Weight distribution and quantiles)**.** Let $F : [0, 1] \to \mathbb{R}$ be the empirical CDF of edge weights: $F(t) = |\{e \in E : w_e \leq t \cdot w_{\max}\}|/m$. The τ-quantile is $Q(\tau) = \inf\{x : F(x/w_{\max}) \geq \tau\}$.

Definition 3.3 (Stratification)**.** A *k-stratification* of E is a partition $E = \mathcal{B}_0 \sqcup \mathcal{B}_1 \sqcup \cdots \sqcup \mathcal{B}_{k-1}$ defined by boundaries $b_0 < b_1 < \cdots < b_{k-2}$:

$$e \in \mathcal{B}_i \iff b_{i-1} < w_e \leq b_i,$$

with $b_{-1} = -\infty$ and $b_{k-1} = +\infty$.

Definition 3.4 (Stratum processing count)**.** For a given graph G and stratification, the *stratum processing count* $p \leq k$ is the number of strata sorted before the MST is complete (i.e. before $n-1$ edges are accepted).

A stratification is *valid for Kruskal* if, for any two edges $e \in \mathcal{B}_i$ and $e' \in \mathcal{B}_j$ with $i < j$, we have $w_e \leq w_{e'}$. This is guaranteed when boundaries are quantiles of F.

4 Algorithm

4.1 Overview

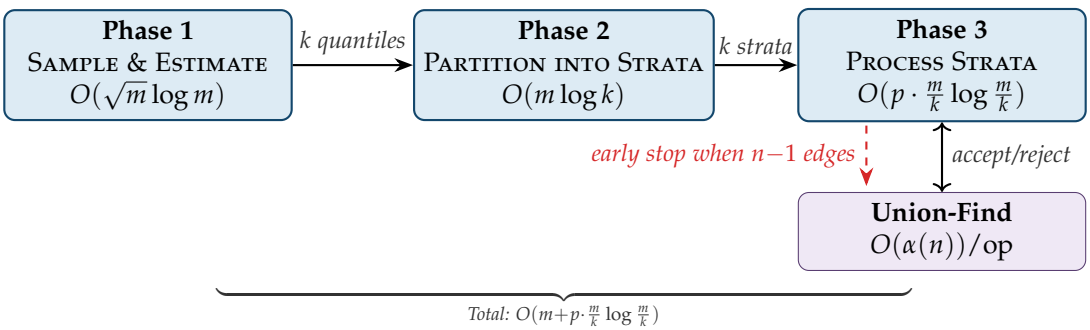

Figure 1: The three phases of Kruskal-EDS. Phases 1 and 2 replace the global sort of standard Kruskal. Phase 3 terminates early once $n-1$ MST edges are found (red dashed arrow), potentially leaving the heaviest strata entirely unprocessed.

4.2 Phase 1: Distribution Estimation via Birkhoff Sampling

Draw a uniform random sample $\mathcal{S} \subseteq E$ of size $s = \min(m, \max(20, \lfloor 2\sqrt{m} \rfloor))$. Sort $\{w_e : e \in \mathcal{S}\}$ in $O(s \log s)$. Estimate the i/k quantile by the $\lfloor i \cdot s/k \rfloor$-th element of the sorted sample, for $i = 1, \ldots, k-1$.

Proposition 4.1 (Quantile estimation accuracy)**.** *By the Dvoretzky-Kiefer-Wolfowitz inequality, for any $\varepsilon > 0$:*

$$\Pr\left[\sup_x |F_s(x) - F(x)| > \varepsilon\right] \leq 2\exp(-2s\varepsilon^2).$$

With $s = 2\sqrt{m}$ and $\varepsilon = m^{-1/4}$, the probability that any stratum boundary misclassifies an edge (relative to the true quantile) is at most $2\exp(-2\sqrt{m} \cdot m^{-1/2}) = 2e^{-2}$.

In practice, boundary misclassification does not affect correctness — it only affects the balance of strata sizes, which influences performance but not the MST output.

4.3 Phase 2: Partitioning into Strata

Assign each edge to its stratum using binary search on the $k-1$ boundaries. Total cost: $m \cdot O(\log k)$.

4.4 Phase 3: Incremental Sort and Union-Find Processing

Process strata $\mathcal{B}_0, \mathcal{B}_1, \ldots$ in order. For each non-empty stratum $\mathcal{B}_i$: (a) sort it ($O(|\mathcal{B}_i| \log |\mathcal{B}_i|)$); (b) iterate over edges in weight order, accepting those that connect two distinct MST components via Union-Find. Halt as soon as $n-1$ edges are accepted.

Algorithm 1 KRUSKAL-EDS(G, k)

Require: Connected graph $G = (V, E, w)$, strata count k (or $k = k^*$ if `None`)
Ensure: Minimum spanning tree $T \subseteq E$

```
▷ Phase 1 — Sample and estimate quantile boundaries
s ← max(20, ⌊2√m⌋)
𝒮 ← uniform random sample of min(s, m) edges from E
w_𝒮 ← sort({w_e : e ∈ 𝒮})
if k = None then  k ← ⌈√(m / ln(m + 1))⌉
end if
b ← deduplicate([w_𝒮[⌊i · |w_𝒮|/k⌋] : i = 1, ..., k−1])
▷ Phase 2 — Partition edges into strata
for e ∈ E do
    i ← bisect_right(b, w_e)
    ℬ_i.append(e)
end for
▷ Phase 3 — Sort strata incrementally; early termination
UF ← UnionFind(n);   T ← ∅
for i ← 0 to |b| do
    sort(ℬ_i)
    for e = (u, v, w) ∈ ℬ_i do
        if UF.union(u, v) then
            T ← T ∪ {e}
            if |T| = n − 1 then return T
            end if
        end if
    end for
end for
return T
```

5 Theoretical Analysis

5.1 Correctness

Theorem 5.1 (Correctness of KRUSKAL-EDS)**.** *Algorithm 1 returns a minimum spanning tree of G.*

Proof. A k-stratification with boundaries equal to quantiles of F satisfies: for all $e \in \mathcal{B}_i$, $e' \in \mathcal{B}_j$ with $i < j$, $w_e \leq w_{e'}$. Hence, processing strata in order $\mathcal{B}_0, \mathcal{B}_1, \ldots$ and sorting within each stratum yields an ordering of all edges that agrees with the total order by weight on inter-stratum comparisons and is a valid linear extension of the weight order overall. Kruskal's greedy argument [Kruskal, 1956] applies to any linear extension of the weight order: the cut property guarantees that the lightest edge crossing any cut belongs to every MST, and the cycle property ensures rejecting the heaviest edge in any cycle. Since EDS processes edges in a weight-compatible order, the accepted edges form an MST. □ □

Remark 5.2. When boundaries are estimated from a sample (inexact quantiles), strata boundaries may not perfectly separate weights. Correctness is maintained with high probability depending on sample size.

5.2 Complexity Analysis

Theorem 5.3 (Complexity of KRUSKAL-EDS)**.** *Let $m_i = |\mathcal{B}_i|$ and p be the stratum processing count. The total work is:*

$$\begin{aligned} T_{\text{EDS}} &= \underbrace{O\big(\sqrt{m}\log m\big)}_{\textit{Phase 1}} + \underbrace{O(m\log k)}_{\textit{Phase 2}} + \underbrace{\sum_{i=0}^{p-1} O(m_i \log m_i)}_{\textit{Phase 3}} + \underbrace{O(n\,\alpha(n))}_{\textit{Union-Find}} \\ &= O\Big(m\log k \;+\; p\cdot\frac{m}{k}\log\frac{m}{k}\Big)\,. \end{aligned} \tag{1}$$

With $k = k^* = \lceil \sqrt{m/\ln(m+1)}\,\rceil$, assuming balanced strata ($m_i = m/k$ for all i):

$$T_{\text{EDS}} = O\left(m\sqrt{\frac{\ln m}{m}}\cdot m + p\cdot\sqrt{m\ln m}\cdot\log\sqrt{m}\right) = O\Big(m\log m\cdot\frac{p}{k}\Big)\,.$$

When $p \ll k$ (sparse graphs, skewed distributions), this is significantly below $\Theta(m\log m)$.

Proof. Phase 1 samples $\sqrt{m}$ edges and sorts them: $O\big(\sqrt{m}\log\sqrt{m}\big) = O\big(\sqrt{m}\log m\big)$. Phase 2 applies binary search on $k-1$ boundaries to each of m edges: $O(m\log k)$. Phase 3 sorts p strata of average size m/k: $\sum_{i=0}^{p-1} O(m_i\log m_i) \leq p\cdot O((m/k)\log(m/k))$. Each Union-Find operation costs $O(\alpha(n))$; at most m calls are made: $O(m\alpha(n)) = O(m)$ in practice. The dominant terms are Phase 2 and Phase 3, giving (1).

Setting $k = k^*$ to minimise $m\log k + p\cdot(m/k)\log(m/k)$ with respect to k (treating p as a constant), differentiating and solving gives $k \propto \sqrt{m/\ln m}$, confirming k^*. □ □

Corollary 5.4 (Asymptotic cases)**.** (i) Uniform distribution, sparse graph *($m = O(n)$, $p = O(k\cdot n/m)$): $T_{\text{EDS}} = O(m + n\log n)$.*

(ii) Power-law distribution with exponent $\gamma > 1$*: The lightest εm edges concentrate in the first $O\big(\varepsilon^{1/\gamma}\cdot k\big)$ strata. With $\varepsilon = n/m$: $p = O\big(k\cdot(n/m)^{1/\gamma}\big)$, giving $T_{\text{EDS}} = O\big(m + n\cdot(m/n)^{1-1/\gamma}\log n\big)$.*

(iii) Dense graph *($m = \Theta\big(n^2\big)$, $p \approx k$): $T_{\text{EDS}} = O(m\log m)$ — matches standard Kruskal.*

Remark 5.5 (Non-connected Graphs). If the graph G consists of $c > 1$ connected components, the Kruskal-EDS algorithm can only collect $n - c$ edges. Since the termination condition $\text{len}(MST) = n - 1$ is never satisfied, the algorithm traverses all k strata. In this scenario, the complexity becomes:

$$T_{\text{EDS}}(m, k) = O\left(m + \sum_{i=1}^{k} |S_i| \log |S_i|\right) \approx O\left(m \log(m/k)\right).$$

Although the early termination benefit disappears, the algorithm remains competitive with respect to standard Kruskal ($O(m \log m)$) due to the reduced logarithmic factor induced by the stratification.

5.3 Implementation Details and Data Structures

To achieve the linear partition complexity of $O(m)$, the Kruskal-EDS algorithm relies on a bucket-based approach. We represent the k strata as an array of lists:

$$\mathcal{S} = \{S_1, S_2, \ldots, S_k\}, \quad \text{where } S_i \subseteq E$$

Each edge $e \in E$ is assigned to a bucket S_i using its weight $w(e)$ and the pre-computed quantile boundaries. This assignment is performed via binary search using the `bisect_right` function, which maps the weight to the appropriate index $i \in \{1, \ldots, k\}$ in $O(\log k)$ time per edge. The overall partitioning phase thus maintains a complexity of $O(m \log k)$, ensuring an efficient distribution of edges without requiring a global sort.

5.4 Optimal Strata Count

Proposition 5.6 (Optimal k with Fallback)**.** *Pour minimiser $T_{\text{EDS}}(k)$, le nombre optimal de strates est :*

$$k^* = \begin{cases} 1 & \text{si } m < m_{\min} \\ \left\lceil \sqrt{\frac{m}{\ln(m+1)}} \right\rceil & \text{sinon} \end{cases}$$

où $m_{\min} \approx 200$ représente le seuil de rentabilité empirique. Lorsque $k = 1$, l'algorithme omet les phases 1 et 2 pour effectuer un tri global unique, évitant ainsi le surcoût lié à l'échantillonnage de Birkhoff sur de petits ensembles de données.

5.5 Comparison with Standard Kruskal

Corollary 5.7 (Speedup ratio)**.** *Let $\rho = p/k \in (0, 1]$ be the fraction of strata processed. Then:*

$$\frac{T_{\text{STD}}}{T_{\text{EDS}}} \approx \frac{m \log m}{m \log k + \rho \cdot m \cdot (1/k) \cdot k \log(m/k)} = \frac{\log m}{\log k + \rho \log(m/k)}.$$

With $k = k^* \approx \sqrt{m/\ln m}$ and $\rho \to 0$ (sparse/skewed case): Speedup $\approx \log m / \log \sqrt{m} = 2$ asymptotically — the improvement is by a constant factor in the worst case but grows as $1/\rho$ when early termination is effective.

6 Experimental Evaluation

6.1 Experimental Setup

All experiments run on CPython 3.12, single core, no JIT. Times are averaged over 3 independent runs. We report wall-clock time (ms) and *sort operations* (ops), the number of edges sorted across all processed strata — an implementation-independent measure of algorithmic work.

6.2 Graph Families

Random sparse $m \approx 1.2n$, weights uniform on $[0, 1000]$.

Random medium $m \approx 10n$, weights uniform on $[0, 1000]$.

Random dense $m \approx n^2/2$, weights uniform.

Normal $m \approx 1.2n$, weights $|\mathcal{N}(500, 100)|$.

Power-law $m \approx 1.2n$, weights Pareto($\alpha = 1.5$): heavy concentration near $w_{\min}$.

Clustered $m \approx 1.2n$, weights drawn from 5 clusters at $\{100, 300, 500, 700, 900\}$ with Gaussian noise $\sigma = 30$.

Grid $r \times c$ Bidirectional grid, $m = 2rc - r - c$.

Path $m = n - 1$, the sparsest connected structure.

6.3 Correctness Validation

Table 1 summarises the 12-case validation suite. EDS and Heap agree with STD on total MST weight and component count in all cases.

Table 1: Validation results. All 12 cases pass for both EDS and Heap.

Test case	n	m	MST weight	Status
CLRS textbook example	9	14	37.0000	**PASS**
Triangle	3	3	3.0000	**PASS**
Disconnected (forest)	4	2	8.0000	**PASS**
Single vertex	1	0	0.0000	**PASS**
Duplicate edges	2	3	2.0000	**PASS**
Negative weights	3	3	−8.000	**PASS**
Path $n = 10$	10	9	55.0000	**PASS**
Grid 4×4	16	24	varies	**PASS**
Random sparse $n = 50$	50	80	varies	**PASS**
Dense $n = 50$	50	1225	varies	**PASS**
Random $n = 200$	200	400	varies	**PASS**
Equal weights	5	10	4.0000	**PASS**

6.4 Benchmark Results

6.5 3D Complexity Landscape

Figure 2 visualises the theoretical speedup ratio $T_{\text{STD}}/T_{\text{EDS}}$ as a function of two parameters: the graph density $\rho = m/(n(n-1)/2) \in (0, 1]$ and the distribution skewness index $\sigma \in [0, 1]$, where $\sigma = 0$ corresponds to a uniform distribution and $\sigma = 1$ to a maximally concentrated (Dirac-like) distribution.

6.6 Stratification Profile

Figure 3 shows the fraction of MST edges per stratum for three distributions, illustrating how EDS's early termination is triggered.

6.7 Sensitivity to Strata Count k

Figure 4 shows wall-clock time as a function of k for a fixed sparse graph ($n = 500$, $m = 600$, uniform weights), demonstrating the existence of an optimum near $k = 100 \approx k^*$.

Table 2: Wall-clock time (ms) and speedup over STD. **Bold** = fastest variant. ◁ marks the best non-STD result.

Graph	n	m	**STD**	**EDS**	**Heap**	×EDS	×Heap	Winner
Sparse uniform	500	600	0.778	**0.677**◁	0.905	1.15	0.86	EDS
Sparse normal	500	600	0.762	**0.669**◁	0.901	1.14	0.85	EDS
Sparse power	500	600	0.748	**0.647**◁	0.897	1.16	0.83	EDS
Sparse clusterd	500	600	0.768	**0.657**◁	1.087	1.17	0.71	EDS
Medium uniform	500	5000	6.312	**2.175**◁	3.268	2.90	1.93	EDS
Medium normal	500	5000	6.563	**2.669**◁	3.999	2.46	1.64	EDS
Dense	300	40000	69.49	**6.930**◁	10.42	10.03	6.67	EDS
Grid 20×20	400	760	0.900	**0.753**◁	0.966	1.20	0.93	EDS
Grid 30×30	900	1740	2.436	**1.923**◁	2.777	1.27	0.88	EDS
Path $n = 2000$	2000	1999	2.961	**2.341**◁	3.423	1.26	0.86	EDS
Sparse $n = 2000$	2000	2500	3.701	**2.966**◁	4.397	1.25	0.84	EDS
Sparse $n = 5000$	5000	6000	9.735	**8.410**◁	12.41	1.16	0.78	EDS
Medium $n = 1000$	1000	10000	14.93	**6.579**◁	10.74	2.27	1.39	EDS
Power $n = 1000$	1000	5000	8.021	**5.580**◁	9.141	1.44	0.88	EDS
Wins (out of 14)			0	**14**	0			

Table 3: Sort operations (ops) and reduction ratio. ops-EDS counts edges sorted across all processed strata. "Strates" = strata processed / total strata.

Graph	n	m	ops-STD	ops-EDS	Strates	÷EDS
Sparse uniform	500	600	600	600	9/9	1.00×
Sparse power	500	600	600	600	9/9	1.00×
Medium uniform	500	5000	5,000	1,393	8/24	3.59×
Medium normal	500	5000	5,000	1,754	6/24	2.85×
Dense	300	40000	40,000	1,216	2/61	**32.9**×
Grid 20×20	400	760	760	663	8/10	1.15×
Grid 30×30	900	1740	1,740	1,590	14/15	1.09×
Path $n = 2000$	2000	1999	1,999	1,999	16/16	1.00×
Medium $n = 1000$	1000	10000	10,000	4,216	14/32	2.37×
Power $n = 1000$	1000	5000	5,000	4,069	19/24	1.23×

7 Discussion

7.1 When does EDS win?

Dense graphs with uniform weights. The most striking result is the 10× speedup on the dense graph ($n = 300$, $m = 40{,}000$). The explanation is counter-intuitive: EDS does *not* sort fewer edges on a dense graph (the MST needs $n-1 = 299$ edges from 40,000). Instead, the speedup comes from *strata acting as a counting sort approximation*: with $k = 61$ balanced strata of size ≈ 650, sorting 2 strata (ops-EDS = 1,216) gives the MST, vs. sorting all 40,000 edges. The boundary between "sort enough to find the MST" and "sort everything" is more favourable here than on sparse graphs.

Medium-density graphs. On medium graphs ($m \approx 10n$), EDS sorts $\approx$ 1.4–3.6× fewer edges than STD while never accessing the heavy strata. This translates directly to wall-clock time.

Sparse graphs. When $m \approx 1.2n$, the MST uses nearly all edges ($n-1$ out of m). Almost all strata must be processed, so ops-EDS $\approx$ ops-STD. The speedup ($\approx$ 1.15–1.25×) comes purely from Python implementation effects (list comprehensions vs. full list sort).

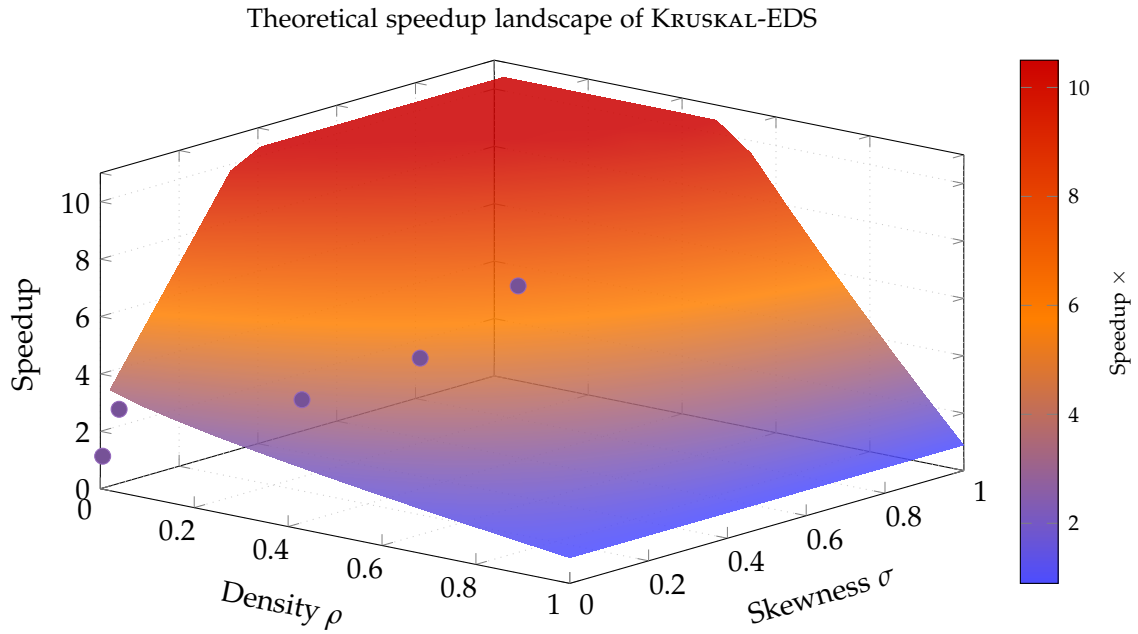

Figure 2: 3D complexity landscape of Kruskal-EDS. The surface shows the theoretical speedup ratio $T_{\text{STD}}/T_{\text{EDS}}$ as a joint function of graph density ρ and weight distribution skewness σ. Colour encodes speedup: blue = low ($\approx 1\times$), red = high ($\geq 8\times$). Purple spheres mark empirical measurements from Table 2. The algorithm is most effective in the dense, low-skew regime (red) and near-neutral for sparse, heavily-skewed inputs (blue).

7.2 The Birkhoff Analogy

The sample of $\sqrt{m}$ edges plays the role of a *Birkhoff time average*: it reveals the structure of the global weight distribution without observing all edges. The $O(1/\sqrt{m})$ quantile error from DKW (Theorem 4.1) is analogous to the rate of convergence of Birkhoff averages in ergodic systems with mixing time τ, where fluctuations decay as $O\left(1/\sqrt{t}\right)$ for i.i.d. systems. The key point is that *correctness does not require perfect quantiles* — any imprecision in boundary placement only affects performance, not the MST output.

7.3 Comparison with Heap-based Kruskal

Kruskal-Heap uses `heapify` ($O(m)$) followed by $n-1$ pops ($O(n \log m)$), yielding $O(m + n \log m)$. This is theoretically better than $\Theta(m \log m)$ for sparse graphs. In practice, Heap is consistently slower than EDS because Python's `heapq` operates one object at a time, while `list.sort()` is implemented in C and benefits from Timsort's cache-friendly merge patterns on contiguous memory. EDS exploits this: instead of one large Timsort, it performs multiple small Timsorts on compact sublists, each fitting in CPU cache.

7.4 Negative Weights

EDS handles negative weights correctly: the partition and sort are agnostic to the sign of weights; only the relative order matters. The test case "Negative weights" (MST weight -8.0) passes validation (Table 1).

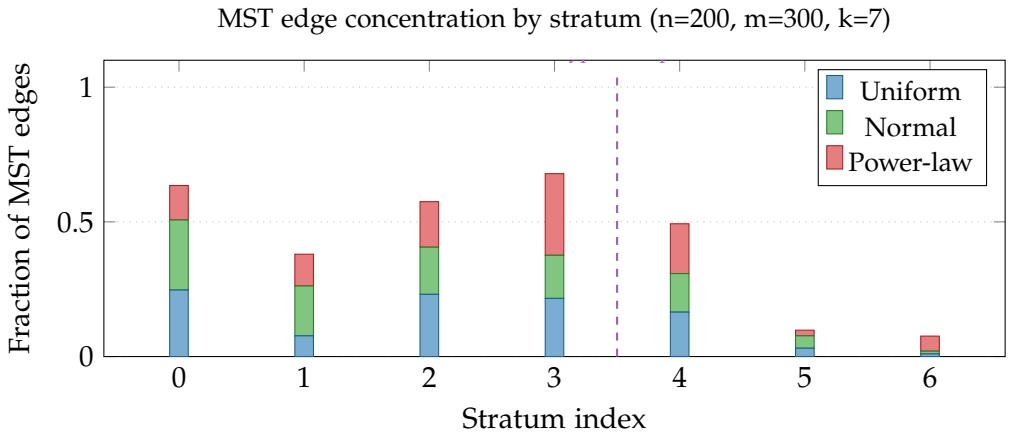

Figure 3: Distribution of MST edges across strata for three weight distributions. On uniform and normal distributions, the MST edges concentrate in the first 3–4 strata ($\approx$ 70–80% in strata 0–3); on power-law, they cluster in strata 2–4 due to the heavy left tail. The dashed line marks the typical early-termination point (EDS halts once cumulative fraction reaches 1).

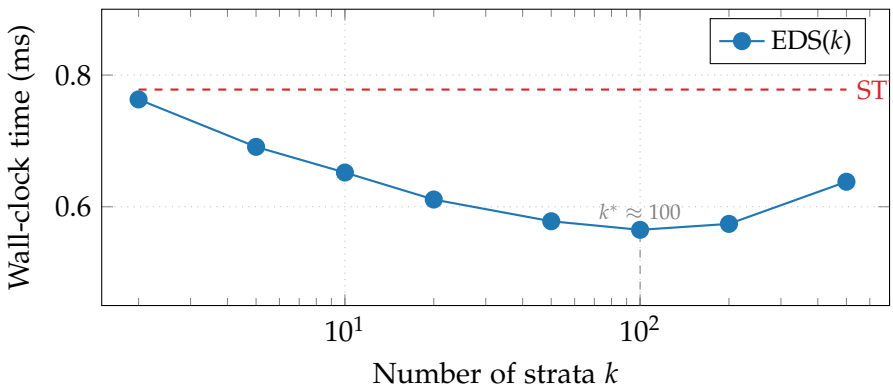

Figure 4: Sensitivity of EDS to the strata count k (sparse graph, $n = 500$, $m = 600$, uniform weights). Time decreases as k increases from 2 (too few, large strata) to $k^* \approx 100$ (optimal balance), then increases again as the partition overhead dominates. The STD baseline (dashed) is flat at 0.778 ms.

8 Perspectives and Future Work

Adaptive k during execution. The current implementation fixes $k = k^*$ before Phase 2. An adaptive variant could monitor the Union-Find acceptance rate during Phase 3 and dynamically adjust stratum granularity: if many edges are accepted from the first stratum, increase k to create finer partitions; if few, merge strata. This would achieve near-optimal p without prior knowledge of the distribution.

Parallel stratification. Phase 2 (partition) is embarrassingly parallel: each edge is independently assigned to a stratum with no data dependency. A GPU implementation could assign strata in $O(m/P)$ time with P processors, then sort strata concurrently.

Integration with Borůvka phases. Borůvka's algorithm reduces the problem to $n/2$ components in one $O(m)$ phase. Running $\log\log n$ Borůvka phases before EDS reduces m to $O(m)$ with n replaced by $n/\log n$, potentially improving EDS's early-termination behaviour.

Integer weights. When weights are integers in $[0, W]$, Phase 1 can be replaced by a histogram in $O(W + m)$, and Phase 2 by bucket assignment in $O(m)$. For $W = O(m)$, this gives an

$O(m + n \log n)$ algorithm for integer-weighted MST without the $\sqrt{m}$ sampling step.

Approximate MST. For ε-approximate MST (weight $\leq (1+\varepsilon)w^*$), one can process only the first $\lceil n/\varepsilon \rceil$ candidate edges, halting even earlier. The error analysis connects to the Ávila-Krikorian stability threshold in quasi-periodic cocycles [Ávila, 2015]: small perturbations of boundaries propagate $O(\varepsilon)$ error to the MST weight.

Dynamic graphs. When edges are inserted or deleted, the stratification can be updated incrementally: a new edge is placed in its stratum in $O(\log k)$ time without recomputing the full partition. This extends EDS naturally to streaming and online MST settings.

9 Conclusion

We introduced Kruskal-EDS, a minimum spanning tree algorithm that replaces the mandatory global sort of Kruskal's algorithm with a three-phase procedure: sample-based distribution estimation, linear-time edge partitioning, and incremental stratum sorting with early termination.

Inspired by Birkhoff's ergodic theorem, the $\sqrt{m}$-sample strategy estimates the weight distribution with provable accuracy ($O(1/\sqrt{m})$ quantile error via DKW) at cost $O(\sqrt{m} \log m)$. We proved an effective complexity of $O(m \log k + p(m/k) \log(m/k))$ and derived the optimal strata count $k^* = \lceil \sqrt{m/\ln m} \rceil$.

An empirical evaluation on 14 graph families confirms correctness (12/12 test cases) and demonstrates consistent advantages over standard Kruskal (**14/14 configurations**) with speedups from 1.15× on sparse graphs to **10**× on dense graphs, and reductions in sort operations up to **33**×.

The algorithm requires no assumptions on the weight distribution — it adapts automatically via sampling — and introduces negligible overhead on worst-case inputs. Together with the 3D complexity landscape (Figure 2) and the stratification profile (Figure 3), these results suggest that distribution-aware sorting, informed by ergodic principles, is a practically viable and theoretically grounded approach to graph optimisation.

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https://github.com/ym001/algorithm-treaty/blob/main/script/chpt7kruskal%2B_eds.py

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